SpecialistRoute Handover — 8. Future Work

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Two checks come first, because they change how existing results are described: the fine-tuning check and a fresh results workbook. After that, the flat 27-class baseline is the one experiment the paper cannot do without.

Do first

- [ ] **Run the fine-tuning check** (Known Issues). If confirmed, relabel those components as frozen-backbone in every document, then decide whether to add two-stage training and retrain them.
- [ ] **Re-run `collect_v2_results.py`.** The last `v2_all_results.xlsx` predates the Potato fix.
- [ ] **Freeze the environment:** `pip freeze > requirements_v2.txt` in `tf_env`.
- [ ] **Confirm look-alike grouping** in the label-quality folds. Every component reported 0 pairs; compare the names in `near_duplicate_review.csv` with the current class folders.

Needed for the paper

- [ ] **Flat 27-class baseline under v2:** same backbone family, 5 seeds, plus its own 5-seed ensemble. Without it the paper cannot claim that routing helps, as opposed to ensembling or backbone choice.
- [ ] **Blind test-set audit:** tag all 236 test images as correct, mislabelled, uninformative or ambiguous, *without* seeing model predictions and with written criteria agreed in advance. Have a second rater tag about 60 images and report Cohen's κ. The official 236 result stays primary; clean-subset accuracy is secondary.
- [ ] **Failure-case figure:** test images that every seed gets wrong, with true vs predicted label, grouped by reason. Nothing is removed from the numbers.
- [ ] **Split the 8.25 pp methodology effect:** retrain one component for one seed while logging test accuracy every epoch (never used for selection). The gap between the best-test epoch and the best-validation epoch is the selection optimism.
- [ ] **Review the label-quality flags:** the 21 Potato images flagged at species level (Classifier sheet) and the 40 lowest-scored Tomato images, tagged y (wrong label) / n (right) / u (can't tell). The u share measures "uninformative image" directly.
- [ ] **Decide on Copy-Paste V2:** regenerate it from the v2 training split (file lookups must respect the split) and re-run the ablation, or drop it from the paper.

Accuracy improvements worth trying

Decide every change on validation (seed 42 first, then 5 seeds) and score test once at the end.

- [ ] **Real fine-tuning,** once two-stage training exists, starting with Tomato (its backbone has always been frozen). Tomato is about half of all errors; +10 pp on Tomato is about +3 pp end to end.
- [ ] **Tomato tweaks:** label smoothing 0.1; a higher input resolution (448–512).
- [ ] **Stronger pretrained features for Tomato:** a single CLIP ViT-L/14 model reaches 77.2% on PlantDoc (PlantWild). First check what Keras 3 can load.
- [ ] **Potato routing:** 4 of 16 Potato test images go to Tomato or Rest. Look at the confusion and at the 21 species-level Potato flags before changing the router.

Already tried, not worth repeating: confidence re-routing between specialists, flip TTA, soft routing, and removing or down-weighting label-quality flags in a linear probe.

Documents to update

These still quote v1 numbers (84.27% and the old per-component values):

- [ ] `PlantDoc_Content_Plan.docx`: rewrite around the methods-led framing and the v2 numbers.
- [ ] `PlantDoc_Assembled_Pipeline_Validation_Process.docx`: v2 protocol and results.
- [ ] Literature comparison (`SpecialistRoute_PlantDoc_Comparison`): drop the state-of-the-art claim; compare with El Karch et al. 2026.
- [ ] `PlantVillage_vs_PlantDoc_Comparison` (table + 2 notes): replace the PlantDoc column with the v2 5-seed values.
- [ ] `CopyPaste_V2_Methodology_Notes.docx`: mark as v1 or update after the Copy-Paste decision.
- [ ] Manuscript skeleton: methods (protocol v2, Potato change, fine-tuning wording) and results sections.